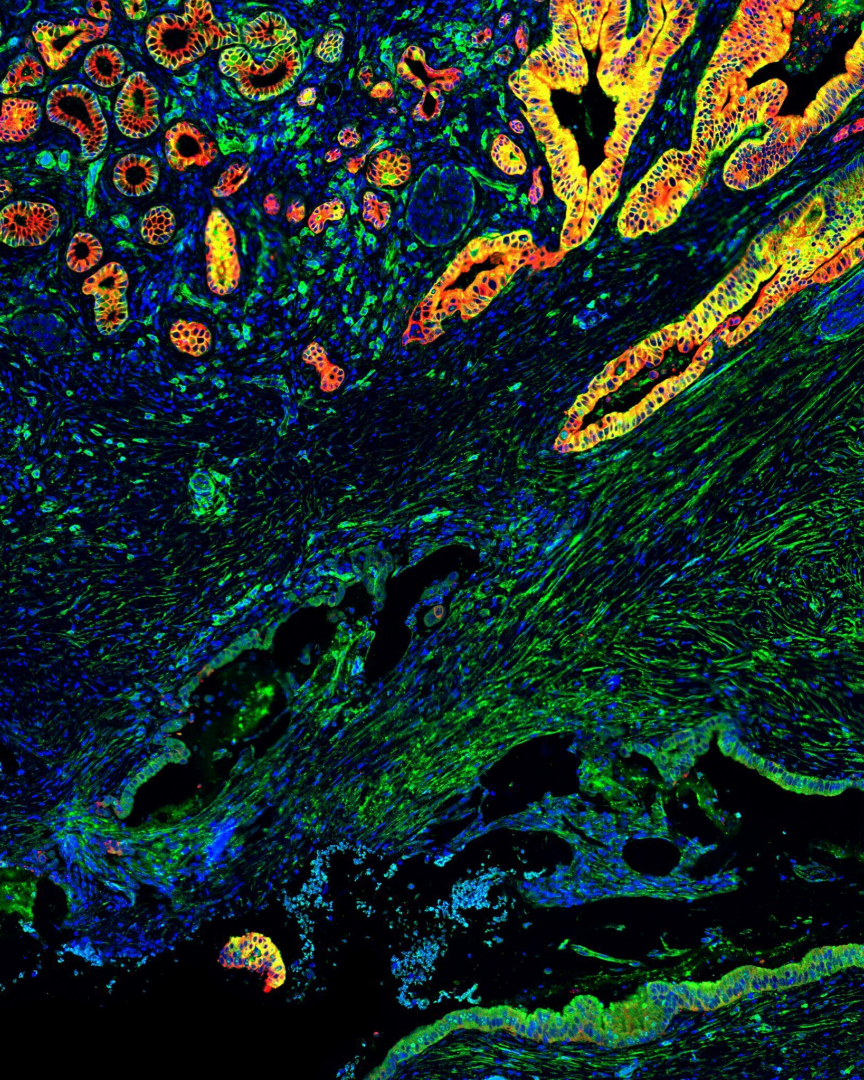


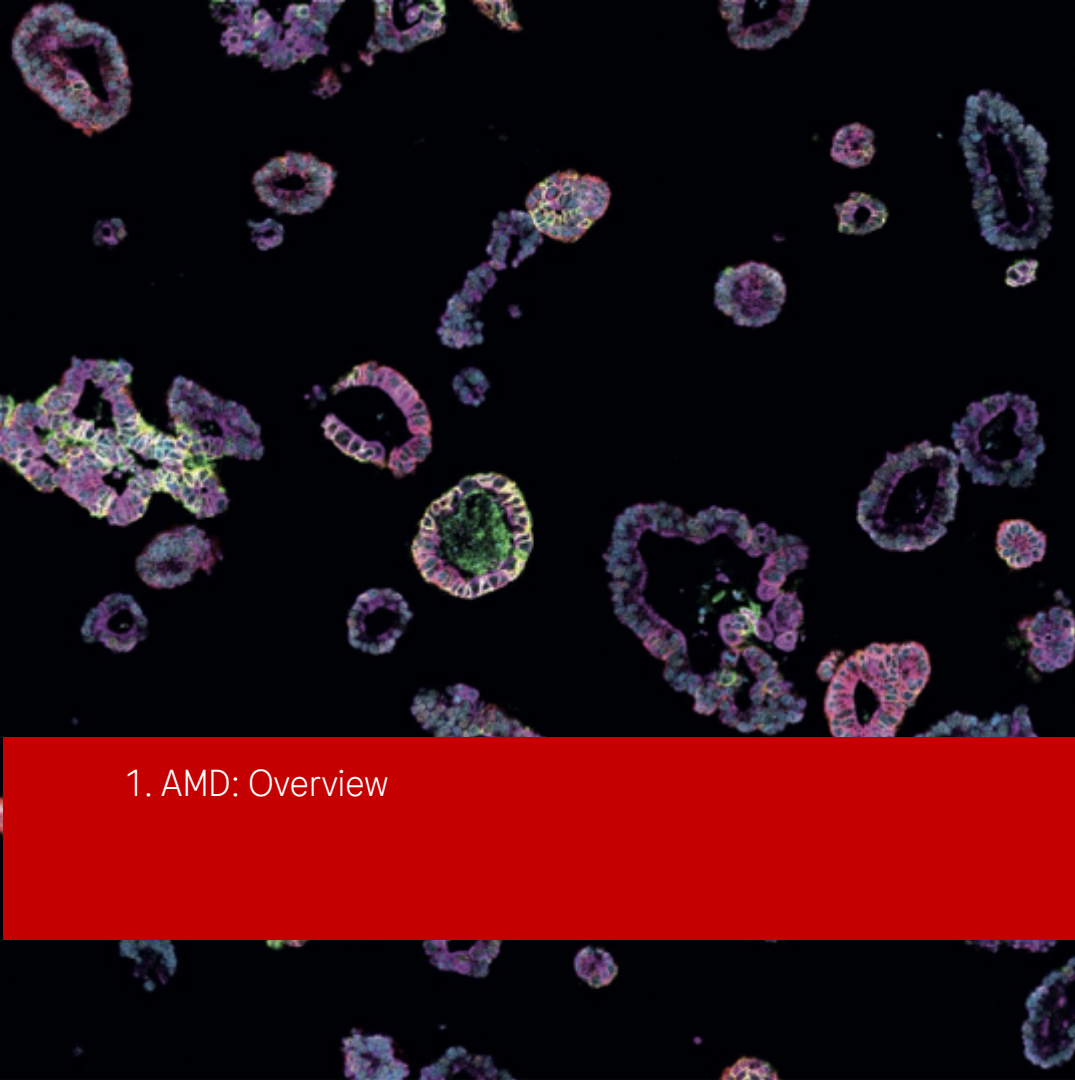


Scoring Tissue Health from Microscopy to Enable Biological Insights in Dry AMD

Sartori Flavio – PhD Intern at IHB
Bo Kyoung Kim – Scientist at IHB

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1. AMD: Overview

Age-Related Macular Degeneration (AMD)



Normal Vision

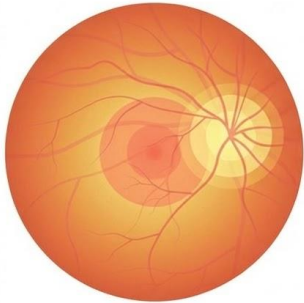


Macular Degeneration

- ❑ What is AMD? It is a common, progressive eye condition (usually) in people over 50 that damages the **macula**, causing **blurry central vision** or **blind spots**, but **rarely total blindness**;
- ❑ Global patient burden continues to rise: **288 million** AMD patients worldwide by 2040.

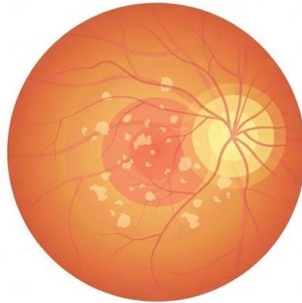
Dry AMD & Wet AMD

NORMAL EYE



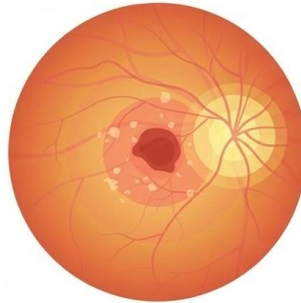
Clear vision of normal eyes

DRY MD

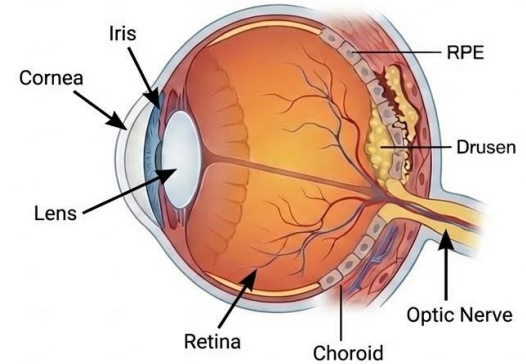


Unusually fuzzy or
distorted vision

WET MD

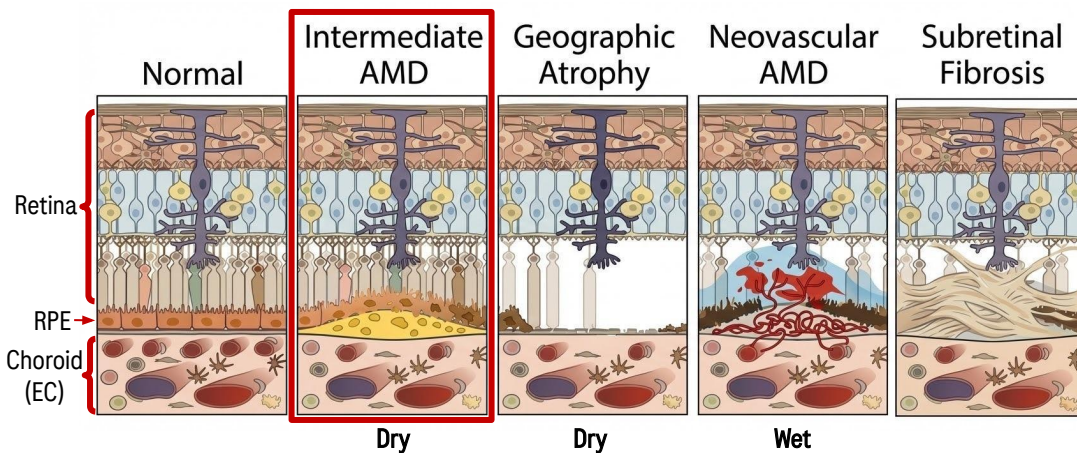


Blind spot in center
of field of vision



- ❑ Dry: gradual loss
- ❑ Wet: rapid, severe leakage
- ❑ Treatments are available for both to slow progression, but there is no cure for late-stage

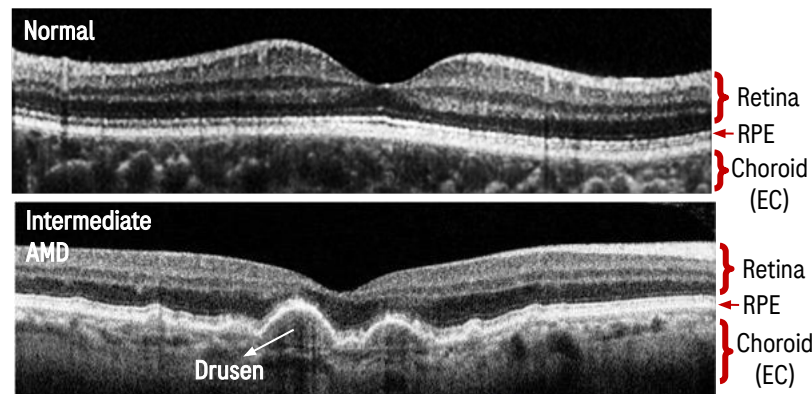
Early AMD pathogenesis begins at the outer blood-retinal barrier



- ❑ AMD progresses through distinct clinical stages with different hallmarks.
- ❑ Intermediate AMD precedes both GA and neovascular AMD.
- ❑ Drusen accumulation is a clinical hallmark.
- ❑ Key pathologies emerge at the outer blood-retinal barrier (oBRB).

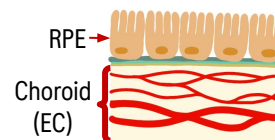
How to diagnose AMD?

Optical coherence tomography (OCT) images



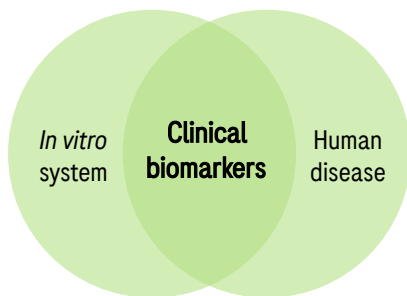
Drusen are yellow deposits under the retina that are made up of lipid and proteins, which are located between the RPE and choroid interface.

Why modeling the oBRB?



oBRB is a key tissue site, where early AMD pathology emerges.

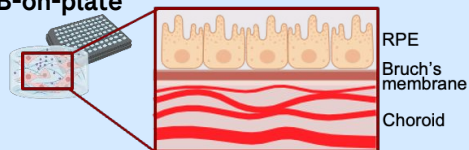
Linking clinical biomarkers to disease pathways using the oBRB model



- Clinical biomarkers serve as **translational anchors** to connect human disease with in vitro phenotype.
- Aim to identify **disease-inducing pathways** that are linked to clinically relevant phenotypes in the oBRB *in vitro* model.

Model development

oBRB-on-plate



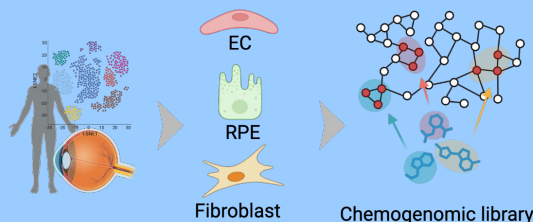
Dissect complex
cell-cell interactions

Complexity
RPE
RPE+EC
RPE+EC+Fibroblast

Reconstruct the human oBRB interface where early AMD-associated biomarkers emerge

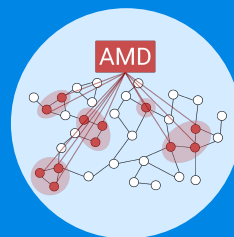
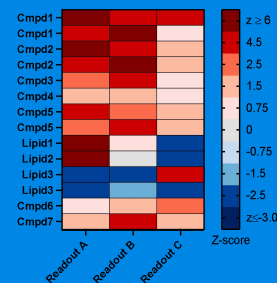
Disease modeling

with clinically anchored phenotypes

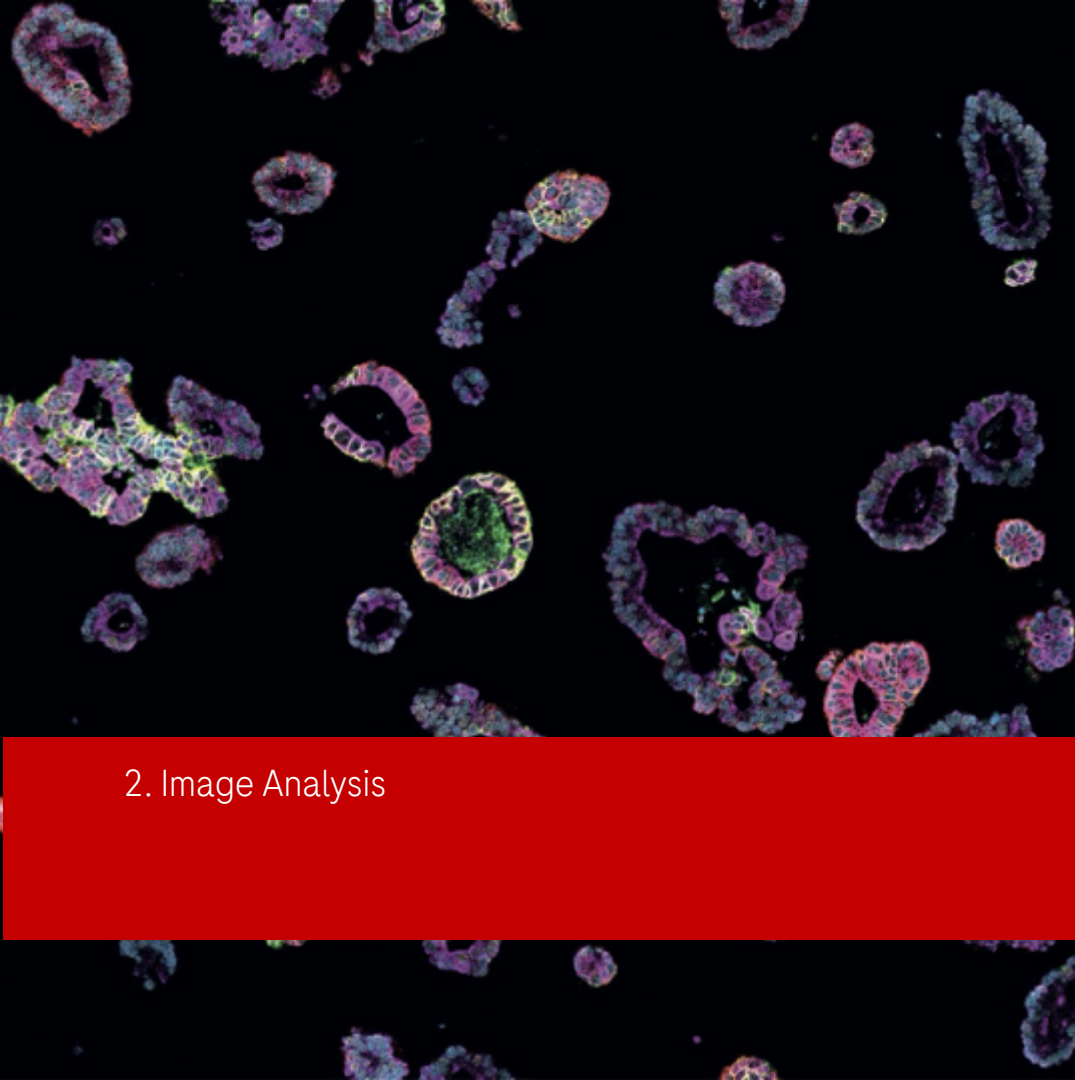


Build a patient-informed chemogenomic library to model clinically anchored phenotypes

Disease pathway identification



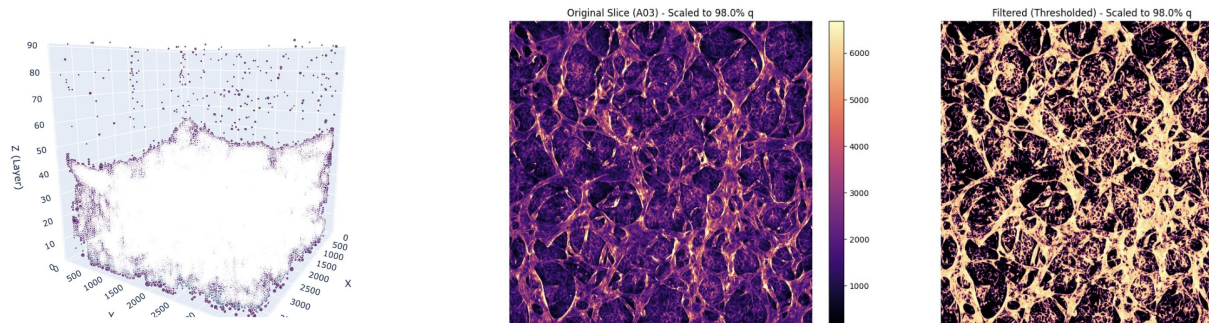
Identify pathways that drive disease-relevant phenotypes in the oBRB model



2. Image Analysis

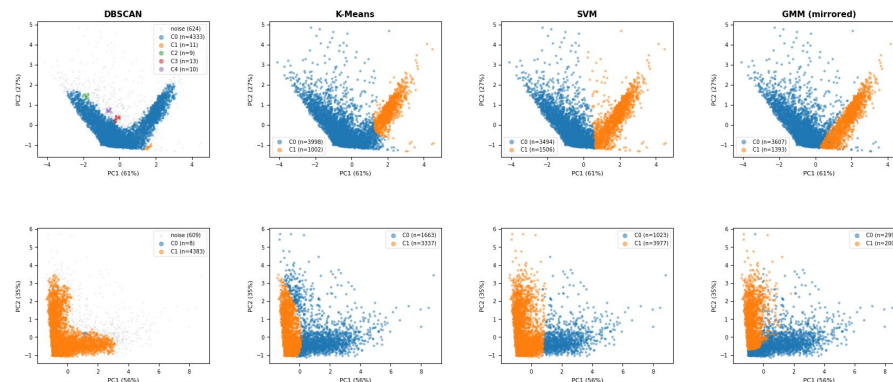
Pre-processing

A bit of AI and Classic techniques



Steps:

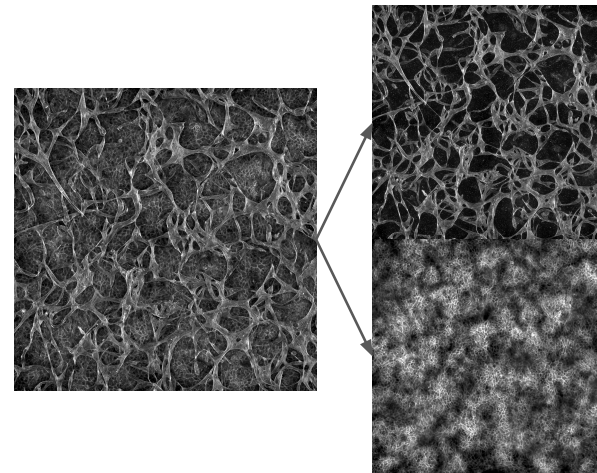
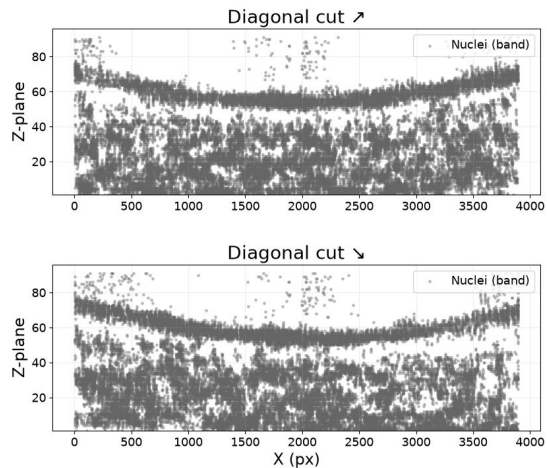
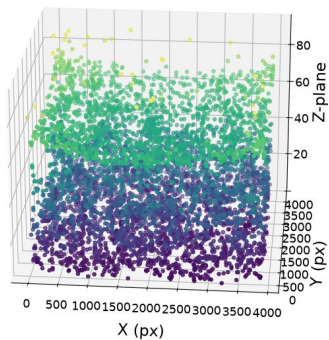
- **3D Reconstruction:** Initial z-stack layer distribution mapped across well field.
- **DAPI Channel & CellPose:** running CellPose on each image to identify the nuclei.



Pre-processing

A bit of AI and Classic techniques

Nuclei — which z is the RPE surface?



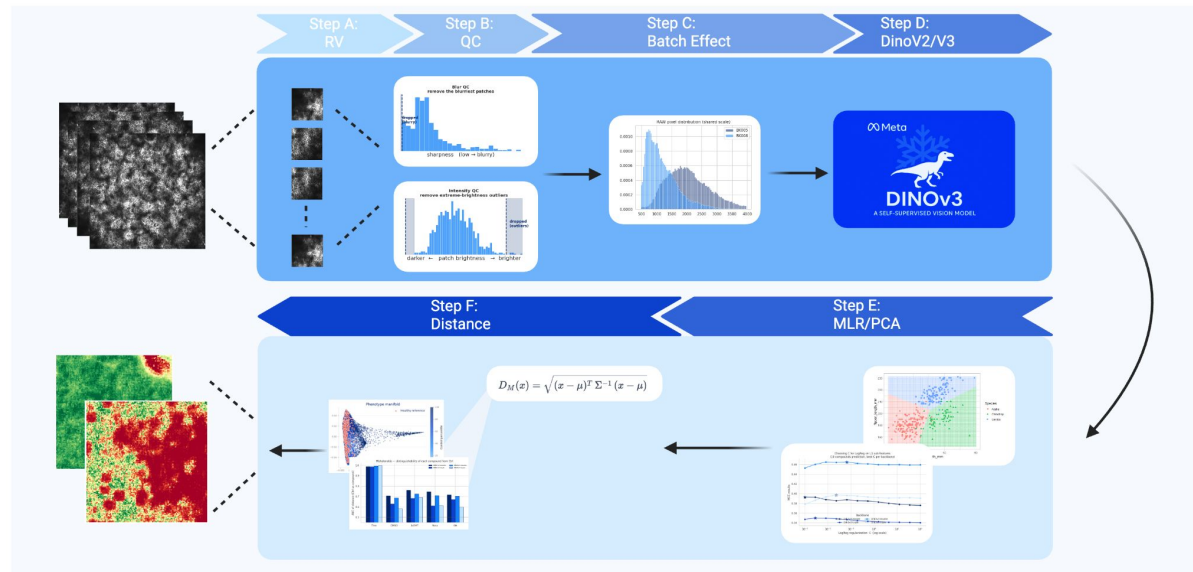
Steps:

- **Classification & Plane Fitting:** GMM model to classify cells and surface fitting to separate the two layers
- **Projection:** ending with two separate projection for above and below layers

Pipeline

Our steps

- Can we extract a health score from a selected patch in a oBRB image?
- Can the health score tell us something beyond what we look in the image?



Steps A & B - Vasculature Removal & Quality Check

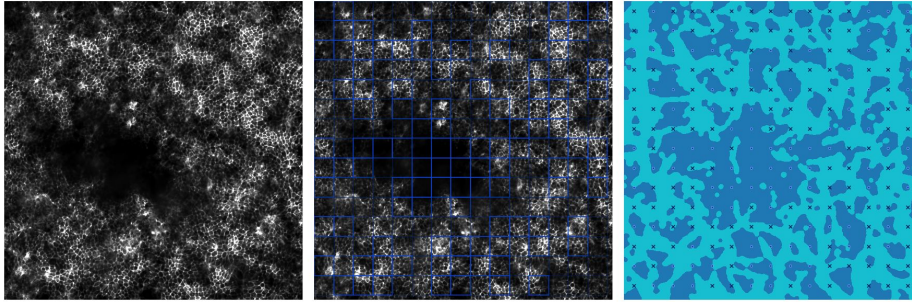
Data exploration & preprocessing

Step A — patch selection: 224x224 grid on the ORIGINAL image + vasculature-center mask

1) Original projection (PROJ_RPE)
251124HBA070228K02_A03_100011901191A01201C03.tif
3900x3900px - grayscale, high dynamic range

2) 224x224 grid, stride 224
kept 101 - dropped 124

3) SEGforeground_labels - (1: 43.2, 2: 56.8) (%area)
drop a patch only if its CENTER is on vasculature



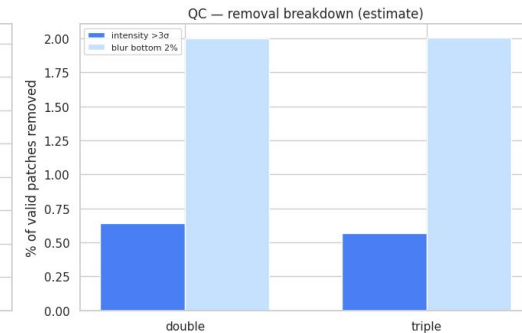
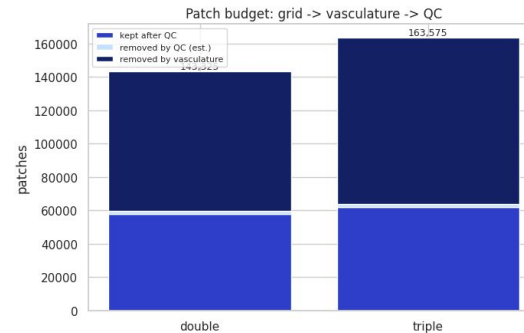
Step A: Vasculature Removal

Removed patches (224x224) centered on vasculature to eliminate scattering artifacts using a skeletonized mask.

- Possible correction with **Monte Carlo Simulation** (?)

Step B: Quality Check

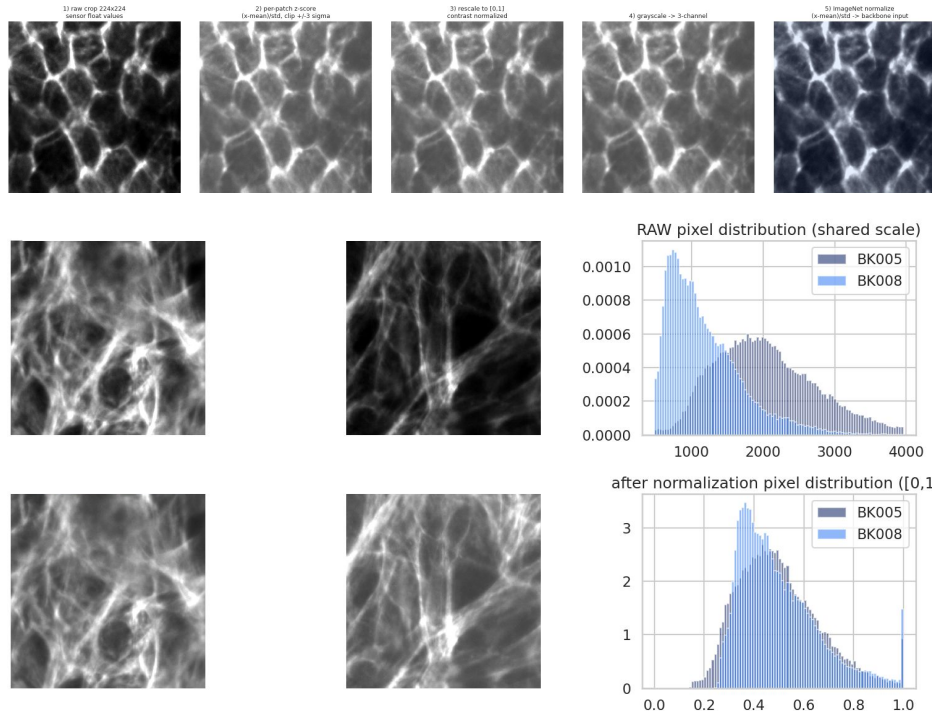
- **Intensity Outliers:** Keep patch if mean intensity is within $\pm 3\sigma$ per well.
- **Variance of Laplacian:** Drop worst 2% tail (blur/focus quality).



Step C - Batch Effect

Shift corrected

Corrected Image Pipeline & Intensity Distribution:



Key Summary

- **Pre-normalization Check:** Evaluated intensity response prior to ImageNet normalization.
- **Intensity Focus:** Primary target is intensity accuracy.
- **Error Correction:** Model requires normalization, but shift errors must be corrected beforehand.

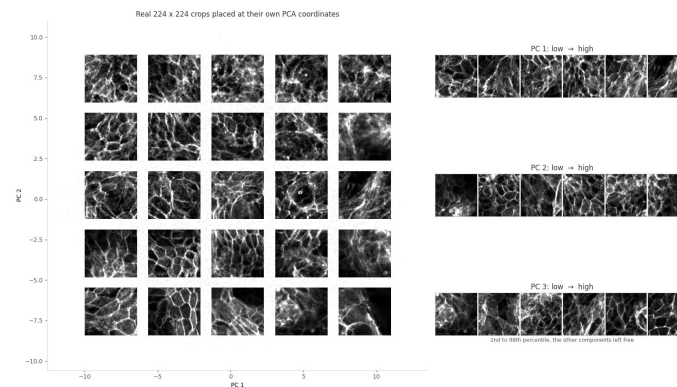
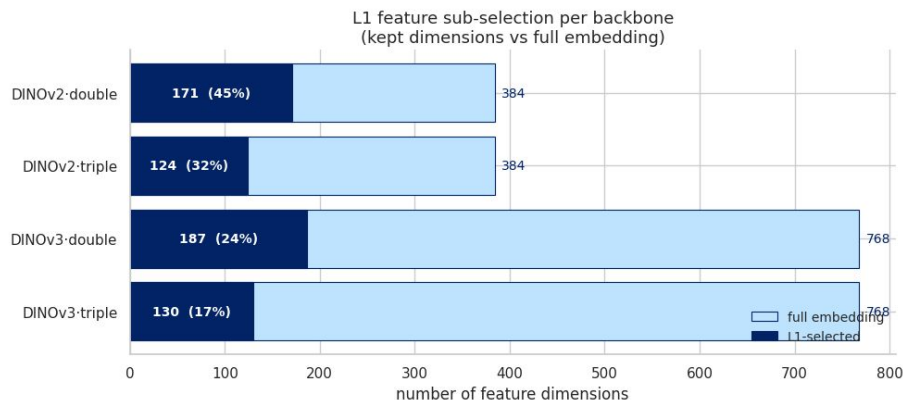


3. Health Score

Step D&E - MLR/PCA

Sub-selecting features

We then fine-tuned **DinoV2** and **DinoV3** on our images. But we ended up with too many features. We can subselect them. The idea is to train a simple classifier (Multinomial Logistic Regression) to classify the different compounds. By applying an L1 Norm with a C small enough to remove more features while keeping performance high enough.



We used different plates of the 16 plates to train the classifier and to obtain a good subsample of features.

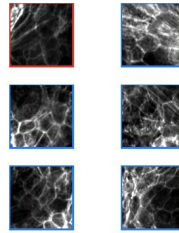
Step F - Health Score → Diffusion Distance

Baseline: **Mahalanobis**

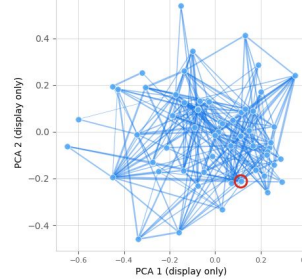
$$D_M(x) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

More advances: **Diffusion**

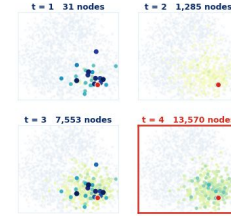
1 Patches become nodes
(red = query, from Compound A)



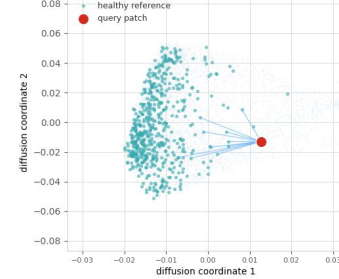
2 Weighted similarity graph
(each node keeps its 30 nearest neighbours)



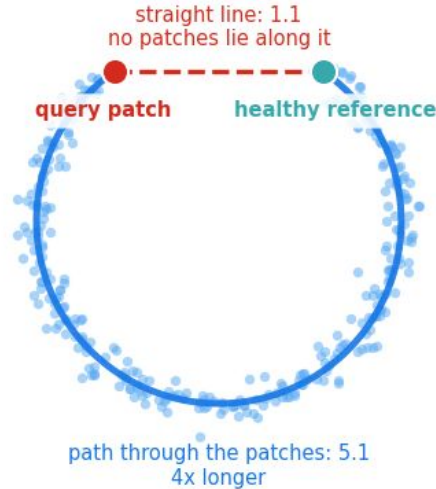
3 Diffusion over four Markov steps
(probability spreads along high-affinity paths)



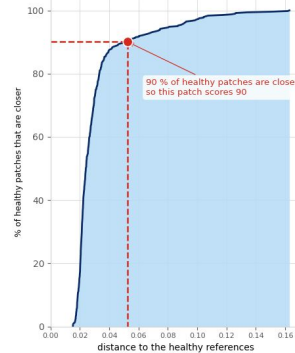
4 Diffusion distance at t = 4
(lines = 10 nearest healthy references)



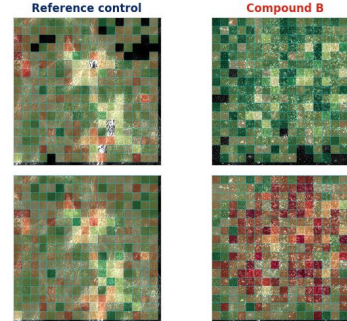
rescaled per snapshot: the peak falls 0.059 to 0.003



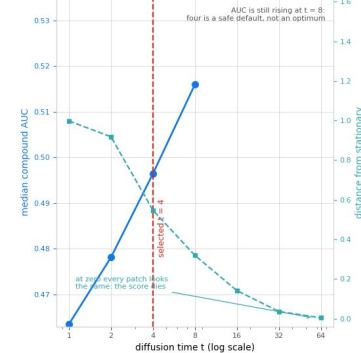
5 From a distance to a 0-100 score
(healthy controls set the ruler)



6 The same two images under both scores

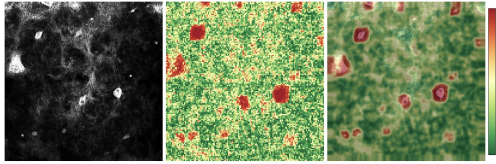


7 Why four steps
(small t keeps the query's identity)



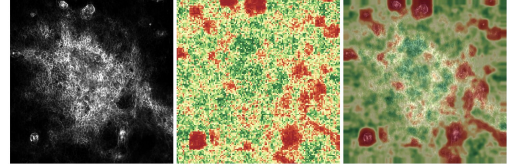
Phenotypic Analysis & Results

Morphological changes across experimental treatment conditions

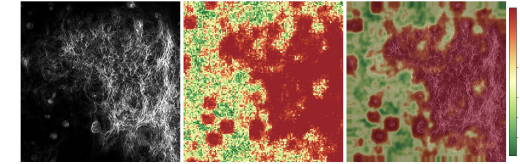


- **Baseline phenotype:** Predominantly green tissue signal as expected for negative control.

- **Artifacts:** Isolated red foci coincide with cell aggregates, edge effects, or background artifacts rather than true biological activity.

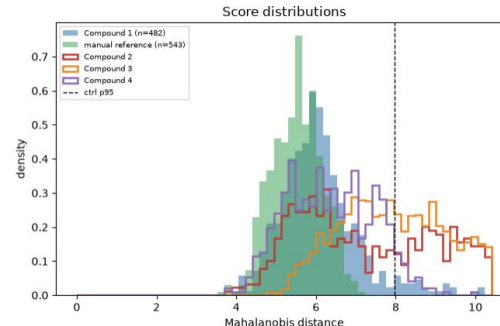


- **Intermediate phenotype:** Intermediate response showing localized tissue remodeling and moderate disruption.



- **Extreme phenotype:** Strongest and most uniform response across the monolayer.

- **Remodeling:** Consistent with widespread disruption (junctional integrity, cell shape, tension, or density).



A fluorescence microscopy image showing various tissue sections, likely from the gastrointestinal tract, stained with multiple fluorescent dyes. The tissues exhibit a range of colors including purple, green, and yellow, indicating different cellular components or markers. The background is black.

4. Future Developments

Summary & Future Developments

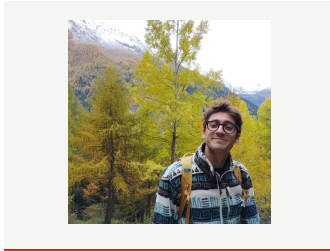
Next steps

Summary:

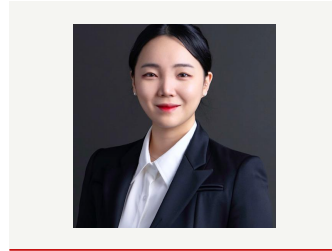
- We built an end-to-end pipeline to extract quantitative information from oBRB microscopy images (3D reconstruction, CellPose-based nuclei detection, GMM-based layer classification and projection)
- We built a health score based on the distance from the control distribution, which led to a consistent ranking of phenotypes – intermediate (localized remodeling) vs. extreme (uniform, monolayer-wide disruption) – while separating genuine biological responses from imaging artifacts

Team Members

IHB



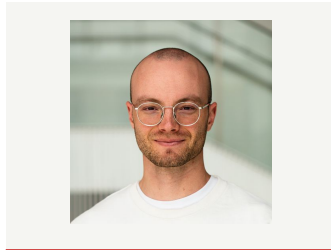
Flavio Sartori
PhD Intern at IHB



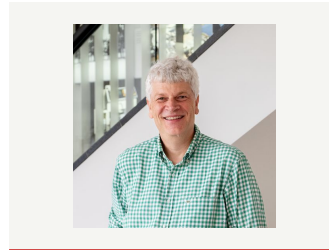
Bo Kyoung Kim
Scientist at IHB



Stephan Hengl
Intern at IHB



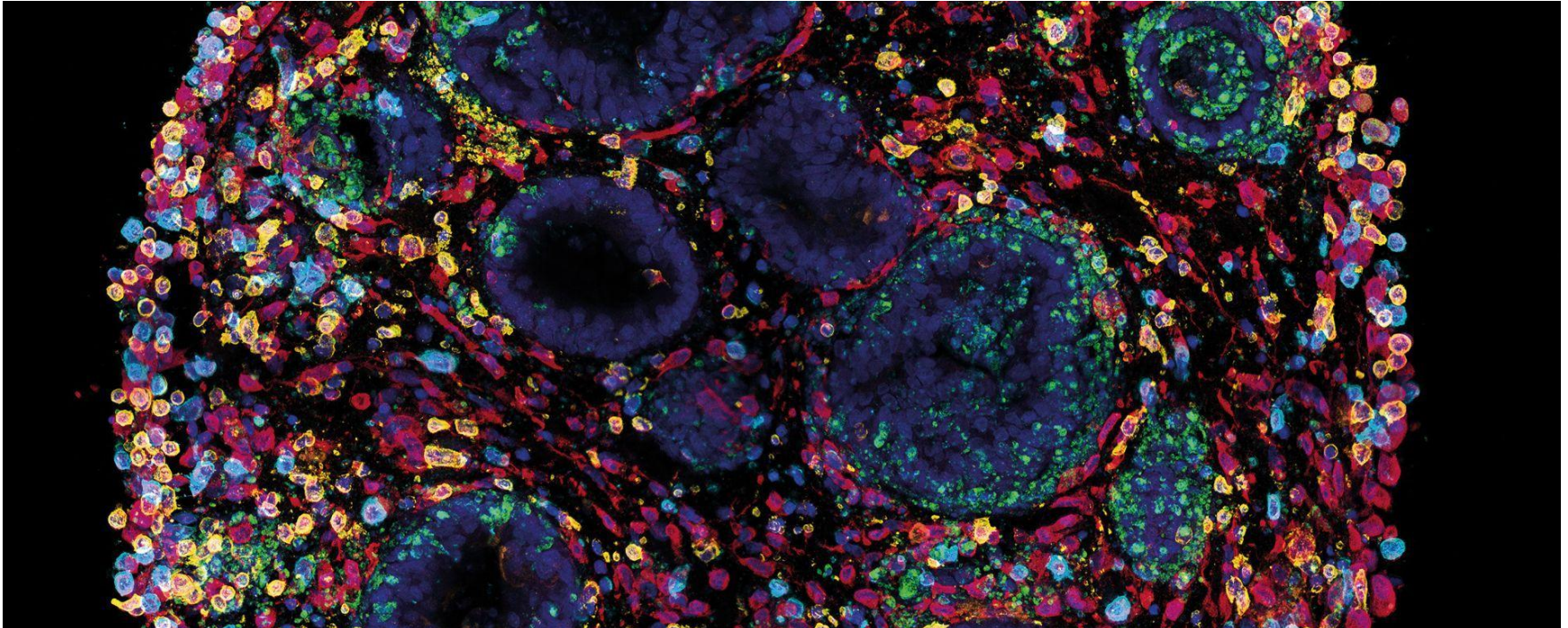
Joris Gentinetta
Software Engineer at IHB



Gaudenz Danuser
Co-Director at IHB

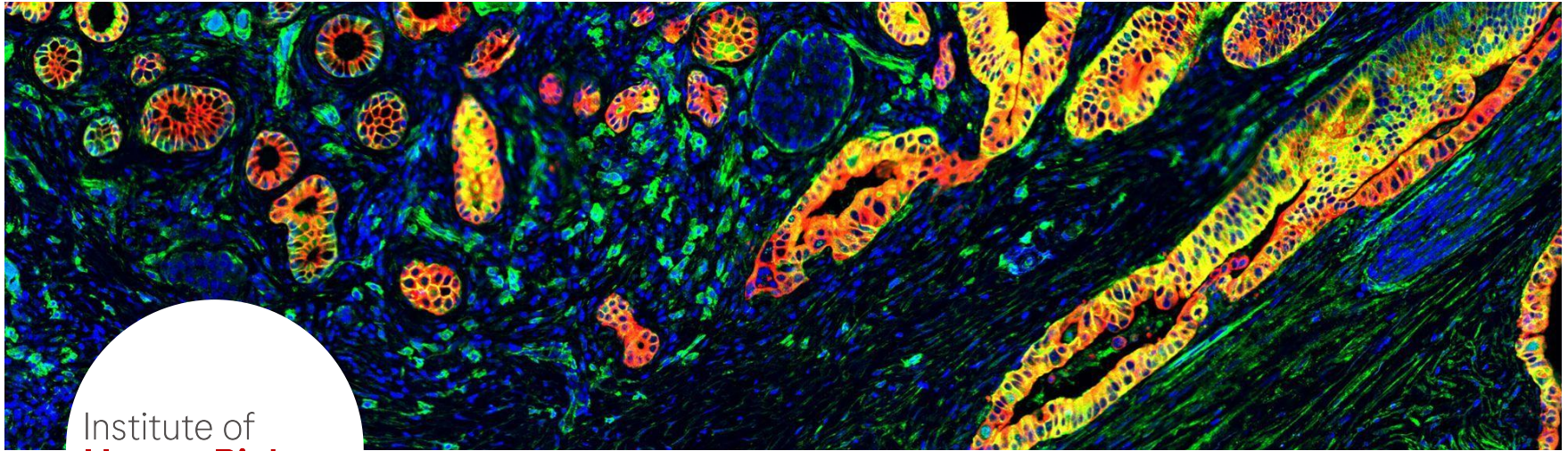


Thank you!



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Doing now what patients need next